

IQ4 Web User Guide

Applies to version v3.50 software



Author: Technical Publications

Issue: 5

Date: 19-Jun-2018

Part Number: TC201256

Copyright: © 2018 Honeywell Products and Solutions SARL, Connected Building Division. All rights reserved.

This manual contains proprietary information that is protected by copyright. No part of this manual may be reproduced, transcribed, stored in a retrieval system, translated into any language or computer language, or transmitted in any form whatsoever without the prior consent of the publisher.

Manufactured for and on behalf of the Connected Building Division of Honeywell Products and Solutions SARL, La Pièce, 16, 1180 Rolle, Switzerland by its Authorized Representative, Trend Control Systems Limited.

For information contact:

Trend Control Systems Limited
St. Mark's Court
North Street
Horsham
West Sussex
RH12 1BW

NOTICE: Trend Control Systems Limited makes no representations or warranties of any kind whatsoever with respect to the contents hereof and specifically disclaims any implied warranties of merchantability or fitness for any particular purpose. Trend Control Systems Limited shall not be liable for any errors contained herein or for incidental or consequential damages in connection with the furnishing, performance or use of this material. Trend Control Systems Limited reserves the right to revise this publication from time to time and make changes in the content hereof without obligation to notify any person of such revisions or changes.

Please send any comments on this or any other Trend technical publication to techpubs@trendcontrols.com

TABLE OF CONTENTS

1	ABOUT THIS MANUAL.....	5
1.1	Conventions Used in this Manual.....	5
1.2	Contacting Trend	6
2	WHAT IS IQ4?	7
3	OVERVIEW OF THE IQ4 WEB INTERFACE	9
3.1	Connect to an IQ4 Controller	9
3.2	Welcome Page	10
3.2.1	Controller Type & Version	10
3.2.2	Title Bar	10
3.2.3	Menu Bar	11
3.2.4	Controller Name, Address & User	11
3.2.5	Navigator Panel.....	11
3.2.6	Information Panel.....	11
3.3	Access Rights (Security)	12
3.4	Refresh the Display	12
3.5	Save New or Changed Settings	12
3.6	Display Layouts on Different Devices.....	13
4	USING THE IQ4 WEB INTERFACE	15
4.1	Login Procedure	15
4.2	Logout Procedure	15
4.3	View and Adjust Occupation Times	16
4.3.1	View Normal Occupation Times.....	17
4.3.2	Change Normal Occupation Times.....	18
4.3.2.1	Copy Occupation Times.....	18
4.3.3	View Exceptions	19
4.3.4	Create or Change Exceptions.....	20
4.3.4.1	Add an Exception.....	20
4.3.4.2	Change an Exception's Parameters.....	21
4.3.4.3	Change an Exception's Times.....	22
4.3.4.4	Copy Settings from an Exception	23
4.3.4.5	Delete an Exception	24
4.4	View and Adjust Module Parameters	25
4.4.1	Adjust the Value of a Knob.....	25
4.4.2	Change the Status of a Switch	26
4.5	Display a Graph	27
4.5.1	Show More or Less Graph Points	27
4.5.2	Display the Graph Data.....	28
4.5.3	Zoom in and Out of a Graph	28
4.6	View Graphical Display Pages (GraphIQs).....	29
4.7	View the Alarm Log	30
5	ADVANCED PROCEDURES.....	31
5.1	View and Adjust Module Parameters (General Guidance).....	31
5.1.1	Navigation Controls (Module List).....	33
5.1.2	View a Connected Module.....	33
5.2	Add a User	34
5.3	Change User Settings.....	35
	INDEX.....	37

1 ABOUT THIS MANUAL

This manual applies to IQ®4 controllers with v3.50 firmware. It provides a description of how to connect to an IQ4 controller web interface and use it to provide basic monitoring and adjustment.

It is assumed that the controller has already been set up and configured, and that the user has knowledge of IQ strategy modules. It is also assumed that the user is familiar with basic computer use and has an understanding of Building Energy Management Systems (BEMS).

The manual is divided into the following sections:

What is IQ4?

This section gives a brief introduction to the IQ4 range of controllers.

Overview of the IQ4 Web Interface

This section provides an introduction to the IQ4 web interface, including how to connect to it, a description of basic navigation features and how it is displayed on a PC or smart device.

Using the IQ4 Web Interface

This section describes how to use the web interface to perform typical day-to-day operations, such as changing occupation times and knob (setpoint) values.

Advanced Procedures

This section covers less commonly used procedures, typically required by system administrators.

1.1 Conventions Used in this Manual

The conventions below are designed to make it quick and easy to find and understand the information.

- The names of text boxes and screens are in **bold** type.
- Options that you need to select are in **bold** type.
- Text you should enter is in *Italic* type.

1.2 Contacting Trend

Head Office

Trend Control Systems Limited
St. Marks Court
North Street
Horsham
West Sussex
RH12 1BW

Tel: +44 (0) 1403 211888

Fax: +44 (0) 1403 241608

Details of regional offices can be found on our web site.

Internet

Our company web site (www.trendcontrols.com) provides information about our products and us. Accredited partners should visit the PNet support site (<https://partners.trendcontrols.com>) for the latest product documentation, firmware downloads and technical support information.

Technical Support

Our support department provides technical support during normal office hours. Before contacting our support department ensure that you have your Technical Support PIN number available, without this we will be unable to provide you with any support.

Tel: +44 (0) 1403 226600

Email: trendts@trendcontrols.com

Fax: +44 (0) 1403 226310

Technical Publications

Please send any comments on this or any other Trend technical publication to techpubs@trendcontrols.com. The latest versions of all Trend technical publications can be obtained from the PNet support site (<https://partners.trendcontrols.com>)

2 WHAT IS IQ4?

The IQ4 range comprises a number of small, highly configurable, intelligent controllers designed for all building control applications. Operating standalone or as part of a network they provide the solution for virtually any type and size of building.

All IQ4 controllers incorporate Ethernet and TCP/IP networking technologies and have an integral web server which can deliver user-specific web pages to a PC or mobile device via a web browser. If a system is set up with the correct TCP/IP connections, a user with the appropriate authorisation can monitor or adjust the controller from any internet access point in the world.

3 OVERVIEW OF THE IQ4 WEB INTERFACE

This section provides an introduction to the IQ4 web interface, including how to connect to it, a description of basic navigation features and how it is displayed on a PC or smart device. For details of how to use the interface to perform specific tasks please refer to [“Using the IQ4 Web Interface” on page 15](#).

The example screen shots included in this manual show typical content and layout when accessed via a web browser on a PC, laptop or large format tablet device. The web interface can also be accessed by smartphones and small format tablets, and on these devices page layouts may differ from those shown. However, the functionality will generally remain the same on all devices. See [“Display Layouts on Different Devices” on page 13](#) for further details.

3.1 Connect to an IQ4 Controller

The web interface of an IQ4 controller can easily be accessed using a web browser, over any TCP/IP network (e.g. company intranet, or the Internet) providing a TCP/IP connection can be made to it. You must also know the controller’s IP Address or Host Name (if the network supports DNS).

Suitable browsers include (but are not limited to) Mozilla Firefox, Internet Explorer, Google Chrome and Safari. Most smartphones and tablets running the latest iOS, Android or Symbian operating systems will work with IQ4 as long as JavaScript is enabled.

Owing to the rapidly changing nature of web devices, new devices and browsers, Trend cannot guarantee a particular device’s compatibility. If you would like to test the compatibility of an IQ4 controller with your browser or device, you can connect to iq4demo.trendcontrols.com and login with username: **mobile** and password: **trend**.

An IQ4 controller can be configured for either HTTP or HTTPS operation.

To connect to an IQ4 controller:

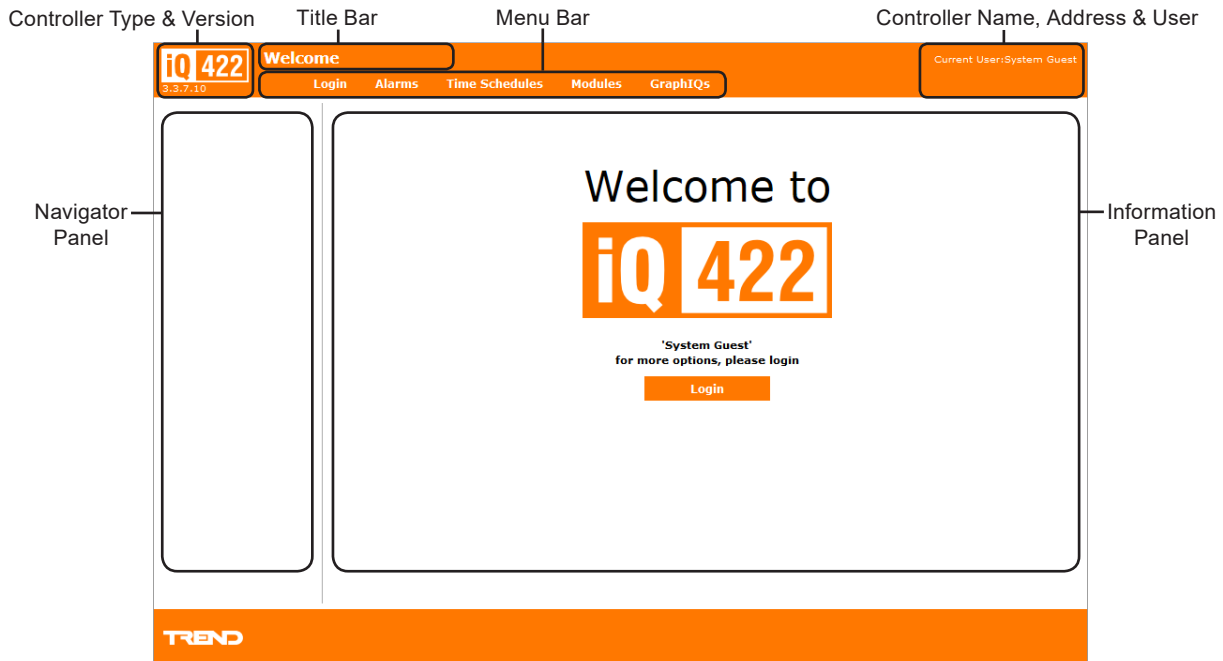
1. Connect to the IP network, and ensure that your network settings are set up to enable a TCP/IP connection to the IQ4 controller.
2. Run your chosen web browser (e.g. Mozilla Firefox).
3. In the address bar of the browser
 - either:* enter the controller’s IP address (e.g. `http://165.195.95.182` or `https://165.195.95.182`),
 - or:* enter the controller’s host name (e.g. `http://iq4mainplant` or `https://iq4mainplant`).

Note: If the controller has been set up to use a TCP/IP port other than port 80, then you will need to specify the port number after the IP address, separated with a colon, e.g. `http://165.195.95.182:6000`.

The IQ4 controller’s Welcome Page will be displayed (see [page 10](#)).

3.2 Welcome Page

When accessing an IQ4 controller from a web browser the first page to be displayed will be the **Welcome Page**. The **Welcome Page** has various features and navigational aids that are common to most other pages:



3.2.1 Controller Type & Version

This area shows the type of controller and its current firmware version. Clicking the icon will return to the Welcome Page from any other page.

3.2.2 Title Bar

This area shows the name of the current page and/or a brief description of its content.

3.2.3 Menu Bar

This area provides links to other top level pages. Depending on your user level and user settings some links may not be displayed.

<i>Button / Link</i>	<i>Function</i>	<i>For further details refer to</i>
	Refresh button. This will request an update from the controller for any displayed values.	“Refresh the Display” on page 12
Login/Logout	Displays a login box allowing a user to enter a username and password. Once logged in the link changes to Logout which will log out the current user.	“Login Procedure” on page 15 “Logout Procedure” on page 15
Alarms	Displays the Alarms Page which lists any alarms in the controller’s alarm log.	“View the Alarm Log” on page 30
Time Schedules	Displays the Time Schedule Overview Page which displays a list of time schedule modules set up in the controller’s strategy. From here it is possible to view and adjust the normal weekly occupation times and any exceptions to those occupation times.	“View and Adjust Occupation Times” on page 16
Modules	Displays the Modules Page which allows you to view all modules set up on the controller and also to view and adjust their parameters.	“Adjust Module Parameters” on page 31
GraphIQs	Displays the top level Graphic Display Page (GraphIQ) configured for the controller, if one exists.	“View Graphical Display Pages (GraphIQs)” on page 29

3.2.4 Controller Name, Address & User

This area shows the name of the controller (i.e. its identifier), its network address and LAN number, and the name of the currently logged-in user.

3.2.5 Navigator Panel

In some pages this area provides additional links to more detailed information.

3.2.6 Information Panel

This area displays details about a selected module or item.

3.3 Access Rights (Security)

Controller security should always be enabled in line with the ‘General Security Best Practice for Trend IP Based Products Information Sheet’ (TP201331). You can login to the web interface using a user name and password that match one of the user modules defined in the controller’s strategy (see [“Login Procedure” on page 15](#)). Once logged in your access rights will be determined by the user module configuration.

However, if security is disabled (i.e. there are no user modules set up) or if security is enabled and you choose not to login then you will be automatically logged in as a ‘casual user’ (i.e. ‘Guest’ or ‘System Guest’) and your access rights will be determined as follows:

<i>Configuration</i>	<i>Casual User Name</i>	<i>User Level</i>	<i>Access Rights</i>
No user modules setup (security is disabled).	‘System Guest’	100	Able to view all information and able to make changes.
‘Guest’ user exists with no password.	‘Guest’	Defined by Guest user module (default = 0)	With the default user level of 0, you can only view the ‘Welcome’ page. Otherwise, access rights will be defined by the Guest user module configuration.
User module(s) setup but ‘Guest’ user does not exist or is set up with password.	‘System Guest’.	-1	Able to view Sensor, Digital Input, Knob, Switch, Driver, Schedule, Time Schedule, Time and Plot modules, the Alarm Log, and GraphIQs but unable to make changes.

For further information refer to [“Add a User” on page 34](#) and [“Change User Settings” on page 35](#).

3.4 Refresh the Display

The data displayed on each page is correct at the time the page is loaded and will not update automatically. To ensure that you are viewing the latest values (e.g. when viewing a graph) you can refresh the display.

To refresh the display:

1. Click  on the menu bar.

3.5 Save New or Changed Settings

Subject to appropriate rights, users can make changes to certain settings on the controller, as described in this manual. Any new settings must be saved (i.e. written to the controller) before they take effect.

If a save operation was successful the following message will briefly be displayed:

Value Write SUCCESS

If a save operation was unsuccessful (e.g. a value was out of range) the following message will briefly be displayed:

Value Write FAILURE

3.6 Display Layouts on Different Devices

The IQ4 web interface is designed to be viewed on a web-enabled PC, laptop, tablet or smartphone.

The layout of each page will change according to the web browser window size and screen resolution of the viewing device. Your choice of web browser may also affect the appearance of the page.

The following examples show how a typical page might appear on different devices:

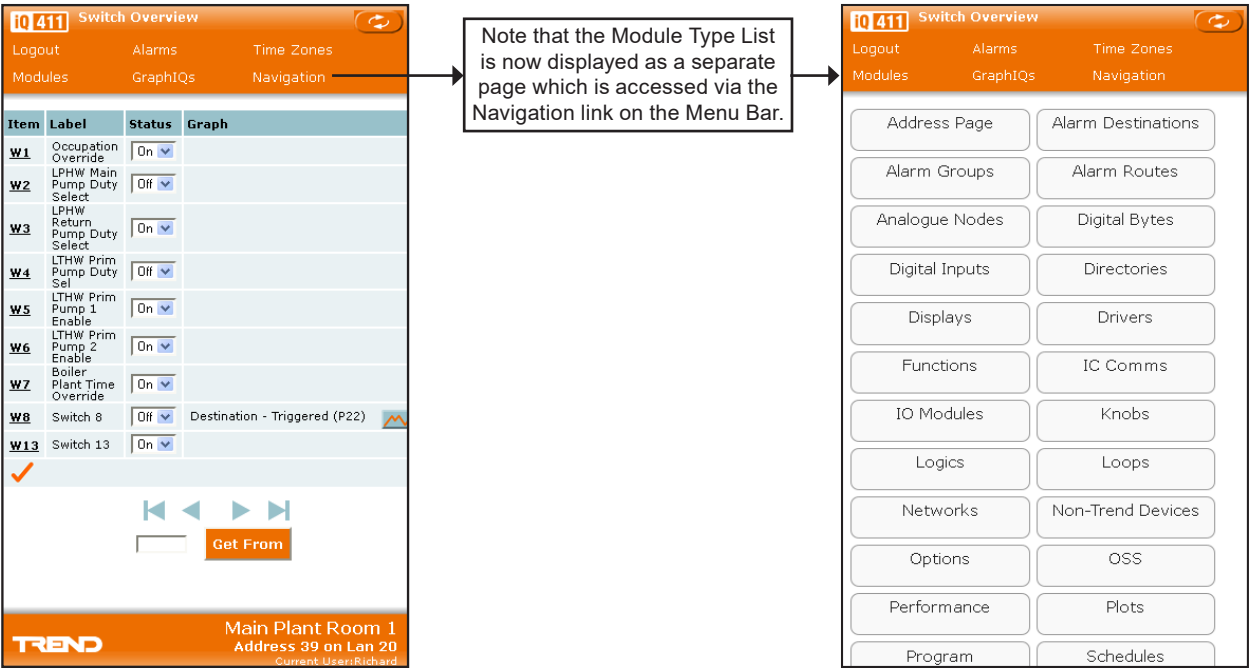
Example 1: As viewed on a PC or laptop

Item	Label	Status	Graph
W1	Occupation Override	On	
W2	LPHW Main Pump Duty Select	Off	
W3	LPHW Return Pump Duty Select	On	
W4	LTHW Prim Pump Duty Sel	Off	
W5	LTHW Prim Pump 1 Enable	On	
W6	LTHW Prim Pump 2 Enable	On	
W7	Boiler Plant Time Override	On	
W8	Switch 8	Off	Destination - Triggered (P22)
W13	Switch 13	On	

Example 2: As viewed on small screen tablet

Item	Label	Status	Graph
W1	Occupation Override	On	
W2	LPHW Main Pump Duty Select	Off	
W3	LPHW Return Pump Duty Select	On	
W4	LTHW Prim Pump Duty Sel	Off	
W5	LTHW Prim Pump 1 Enable	On	
W6	LTHW Prim Pump 2 Enable	On	
W7	Boiler Plant Time Override	On	
W8	Switch 8	Off	Destination - Triggered (P22)
W13	Switch 13	On	

Example 3: As viewed on a smartphone



4 USING THE IQ4 WEB INTERFACE

This section describes how to use the web interface to perform typical day-to-day operations, such as changing occupation times and knob (setpoint) values.

Note: The example screen shots in this manual show typical content and layout when accessed via a web browser on a PC, laptop or large format tablet device. When accessed by smartphones and small format tablets the page layouts may differ from those shown. However, the functionality will generally remain the same on all devices. For examples of how a page may appear on different device types, see [“Display Layouts on Different Devices” on page 13](#).

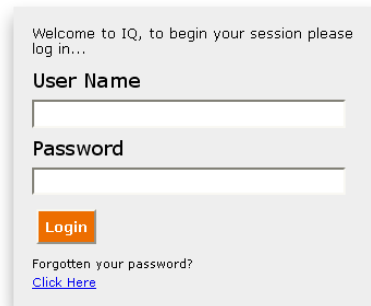
4.1 Login Procedure

In order to login to the controller you must enter a username and password that matches one of the user modules set up in the controller.

If no user modules have been set up in the controller you will not be able to log in and you will be given either ‘Special Guest’ or ‘Guest’ status (see [“Access Rights \(Security\)” on page 12](#)).

To login:

1. Connect to the IQ4 Controller (see [“Connect to an IQ4 Controller” on page 9](#)).
2. Click **Login** on the menu bar. The login box will appear:



3. In the **User Name** box enter your user name.
4. In the **Password** box enter your password.
5. Click the **Login** button.

If you are successfully logged in your user name will appear as the Current User in the top right corner of the page. If a specific home page has been configured this will be displayed, otherwise the **Welcome page** will still be displayed.

Note: Your access rights to view or change information is defined by the security settings of the user module.

4.2 Logout Procedure

Once you have finished using the IQ4 web interface, you should always logout to prevent any unauthorised changes being made. You will also be logged out automatically after a period of inactivity if your user profile has been configured with a timeout.

To log out manually:

1. Click **Logout** on the menu bar.
2. Close the browser.

4.3 View and Adjust Occupation Times

Normal week occupation times specify daily on/off periods that occur every week. **Exceptions** provide a means of overriding the normal occupations times for specific dates (e.g. bank holidays or shutdown periods).

Normal occupation times and exceptions are contained within time schedules and these are viewed or adjusted from the Time Schedule Overview.

To view the Time Schedule Overview:

1. Login to the IQ4 Controller (see [“Login Procedure” on page 15](#)).
2. Click **Time Schedules** on the menu bar. The Time Schedule Overview will be displayed:

IQ 422 Time Schedule Overview **Tech Pubs IQ422**
3.3.7.10 Address 16 on Lan 18 Current User: Admin

Logout Alarms Time Schedules Modules GraphIQs

Zone	Label	Value	Normal Week	Exceptions
Z1	Office Times	Off	Normal Week	Exceptions
Z2	Reception Times	Off	Normal Week	Exceptions

Navigation: [Back] [Forward] [Search] [Get From]

TREND

Note: The Time Schedule Overview can also be accessed by selecting the Time Schedules module type on the Modules Page (see [page 31](#)).

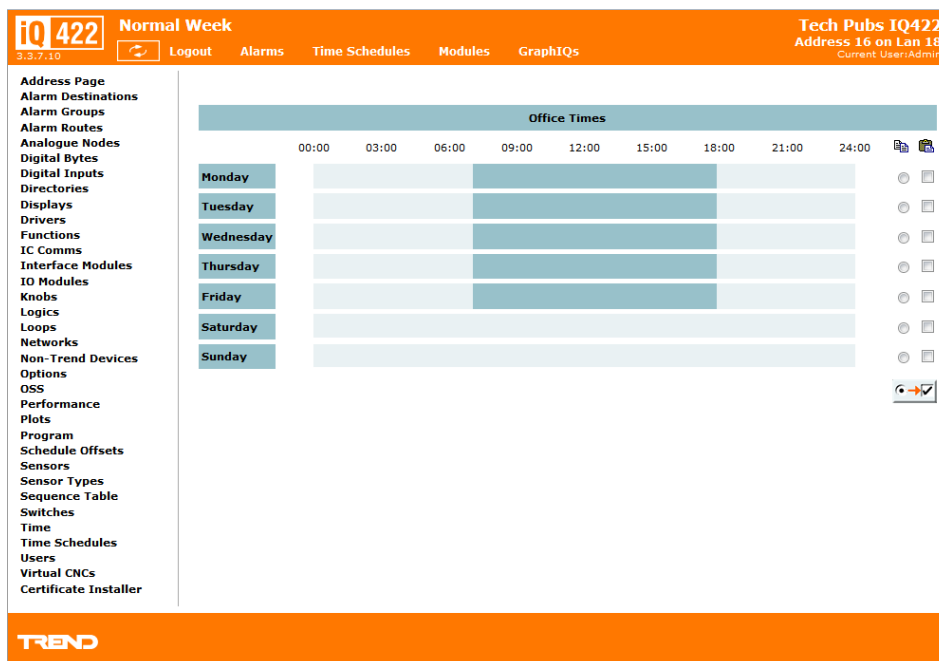
Up to 25 time schedules will be listed. If there are more than 25 time schedules, use the navigation controls to change the list view (see [page 33](#)).

4.3.1 View Normal Occupation Times

Normal occupation times specify the occupation times that will be used each day, unless these are overridden by an exception.

To view the normal occupation times:

1. View the Time Schedule Overview Page (see [page 16](#)).
2. Click **Normal Week** for the schedule that you want to view. The Normal Week Page will be displayed:



This indicates the occupation times for each day of the week. To view actual on/off times for a particular day click the required day name (e.g. **Monday**):

3. To change the occupation times, see [“Change Normal Occupation Times” on page 18](#).

4.3.2 Change Normal Occupation Times

Normal occupation times can be changed provided that you have a user level that equals or exceeds the level set for the time schedule.

You can edit the existing times for each day, add or delete occupation periods, and copy the times from one day to one or more other days.

To change the normal occupation times:

1. View the normal occupation times (see [“View Normal Occupation Times” on page 17](#)).
2. Click the day for which the occupation times are to be changed. The times for the chosen day will be displayed:

3. In the box in the **Time** column enter the required start (on) or stop (off) time, using the 24-hour format hours:minutes (e.g. 17:30 for 5.30pm). To change a time from a start time to a stop time or from stop time to a start time, select **On** or **Off** in the **Value** column.
4. To add extra time periods click **Create New Time Schedule Period Module**. Another period will be added to the list. Enter the **Time**, and specify whether it is a start (on) or stop (off) time.

Note: New periods should be always added in pairs, one to switch occupation on, and one to switch it off.

5. To delete a time period, click . The controller will prompt for confirmation, click **OK**.
6. Click to save the new occupation times.
7. Repeat for any other period that requires changing.

Hint: The occupation times for an entire day can be copied to other days as described below.

4.3.2.1 Copy Occupation Times

Occupation times for one day in a Normal Week can be copied to one or more other days.

To copy occupation times:

1. View the normal occupation times (see [“View Normal Occupation Times” on page 17](#)).
2. Select the radio button in the column next to the day whose occupation times are to be copied.
3. Select the check box(es) in the column next to the day(s) to which the times are to be pasted.
4. Click to copy the times to the selected day(s).

4.3.3 View Exceptions

Exceptions are used to define different occupation times to be applied for specific dates or periods.

To view exceptions:

1. View the Time Schedule Overview Page (see [page 16](#)).
2. Click **Exceptions** for the schedule that you want to view. The Exception Profile Overview Page is displayed containing a list of any exceptions:

Exception Profile Overview

Tech Pubs IQ422
Address 16 on Lan 18
Current User: Admin

Logout Alarms Time Schedules Modules GraphIQs

Address Page
Alarm Destinations
Alarm Groups
Alarm Routes
Analogue Nodes
Digital Bytes
Digital Inputs
Directories
Displays
Drivers
Functions
IC Comms
Interface Modules
IO Modules
Knobs
Logics
Loops
Networks
Non-Trend Devices
Options
OSS
Performance
Plots
Program
Schedule Offsets
Sensors
Sensor Types
Sequence Table
Switches
Time
Time Schedules
Users
Virtual CNCs
Certificate Installer

	Start Day	Start Month	End Day	End Month	Priority	Use	Delete	Times
New Year Holiday	1	Jan	1	Jan	1	Every		
Easter Holiday	16	Apr	19	Apr	1	Next		
Bank Holiday	25	May	25	May	1	Next		
Summer Shutdown	1	Aug	16	Aug	1	Next		

Create New Exception Profile

TREND

This list shows the dates to which the exceptions apply, their priority, and whether they occur only once, or every year.

3. To view the occupation times for an exception click  next to the required exception. The Day Profile Overview is displayed:

Day Profile Overview

Tech Pubs IQ422
Address 16 on Lan 18
Current User: Admin

Logout Alarms Time Schedules Modules GraphIQs

Address Page
Alarm Destinations
Alarm Groups
Alarm Routes
Analogue Nodes
Digital Bytes
Digital Inputs
Directories
Displays
Drivers
Functions
IC Comms
Interface Modules
IO Modules
Knobs
Logics
Loops
Networks
Non-Trend Devices

Office Times

Bank Holiday

00:00 03:00 06:00 09:00 12:00 15:00 18:00 21:00 24:00

Times

Note: Clicking the time schedule name at the top of page (e.g. Office Times) will return to the list of exceptions.

4. To edit exceptions, see [“Create or Change Exceptions” on page 20](#).

4.3.4 Create or Change Exceptions

You can add a new exception, or copy, modify or delete an existing exception, provided that you have a user level that equals or exceeds the level set for the time schedule.

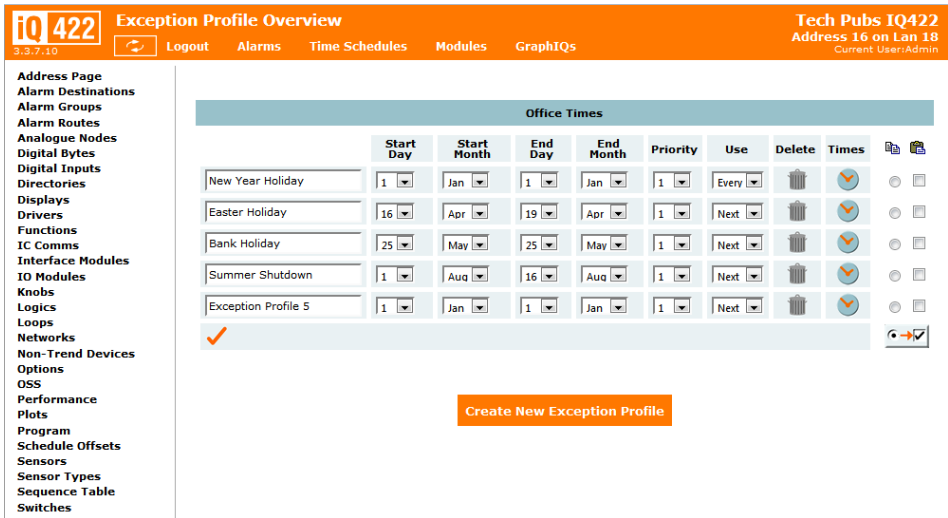
4.3.4.1 Add an Exception

New exceptions can be added as required.

Note: It is good practice to delete old or unused exceptions to maintain free memory space in the controller (see [“Delete an Exception” on page 24](#)).

To add an exception:

1. View the exceptions (see [“View Exceptions” on page 19](#)).
2. Click **Create New Exception Profile**. The exception will be added to the list using various default values:



3. Set up the exception's parameters as described in [“Change an Exception's Parameters” on page 21](#).
4. Set up the exception's occupation times as described in [“Change an Exception's Times” on page 22](#).

4.3.4.2 Change an Exception's Parameters

An exception's parameters specify on what date (or date range) the exception is to occur, whether it is to be repeated every year, and its priority.

To set up an exception's parameters:

1. View the exceptions (see [“View Exceptions” on page 19](#)).

	Start Day	Start Month	End Day	End Month	Priority	Use	Delete	Times
New Year Holiday	1	Jan	1	Jan	1	Every		
Easter Holiday	16	Apr	19	Apr	1	Next		
Bank Holiday	25	May	25	May	1	Next		
Summer Shutdown	1	Aug	16	Aug	1	Next		
Exception Profile 5	1	Jan	1	Jan	1	Next		

[Create New Exception Profile](#)

2. Enter a label for the exception in the left most box.
3. In the **Start Day** box select the day that the exception times are to be first used.
4. In the **Start Month** box select the month that the exception times are to be first used.
5. In the **Stop Day** box select the last day that the exception times are to be used.
6. In the **Stop Month** box select the month for the exception is to end.
7. In the **Priority** box select the priority for the exception.

The priority parameter is used when two exceptions happen on the same day. The exception with the lower value priority will take precedence, e.g. priority of 1 takes precedence over 3. If there are exceptions with the same priority the exception with the earliest start time takes precedence.

8. In the **Use** box select the required option to specify when the exception is to occur:

Every - the exception will be repeated every year,

Next - the exception will occur only once,

Free - the exception won't be used.

9. Click to send the changes to the controller.

The exception's times must be set up separately as described in [“Change an Exception's Times” on page 22](#).

4.3.4.3 Change an Exception's Times

Once an exception has been added its occupation times must be set up.

To set up an exception's times:

1. View the exceptions (see [“View Exceptions” on page 19](#)).

Exception Profile Overview

Tech Pubs IQ422
Address 16 on Lan 18
Current User: Admin

Logout Alarms Time Schedules Modules GraphIQs

Address Page
Alarm Destinations
Alarm Groups
Alarm Routes
Analogue Nodes
Digital Bytes
Digital Inputs
Directories
Displays
Drivers
Functions
IC Comms
Interface Modules
IO Modules
Knobs
Loops
Networks
Non-Trend Devices
Options
OSS
Performance
Plots
Program
Schedule Offsets
Sensors
Sensor Types
Sequence Table
Switches

	Start Day	Start Month	End Day	End Month	Priority	Use	Delete	Times
New Year Holiday	1	Jan	1	Jan	1	Every		
Easter Holiday	16	Apr	19	Apr	1	Next		
Bank Holiday	25	May	25	May	1	Next		
Summer Shutdown	1	Aug	16	Aug	1	Next		
Exception Profile 5	1	Jan	1	Jan	1	Next		

Create New Exception Profile

2. Click the icon for the required exception. The Day Profile Overview will appear showing the occupation times (if any) for the exception:

Day Profile Overview

Tech Pubs IQ422
Address 16 on Lan 18
Current User: Admin

Logout Alarms Time Schedules Modules GraphIQs

Address Page
Alarm Destinations
Alarm Groups
Alarm Routes
Analogue Nodes
Digital Bytes
Digital Inputs
Directories
Displays
Drivers
Functions
IC Comms
Interface Modules

Office Times

Open Day Event

00:00 03:00 06:00 09:00 12:00 15:00 18:00 21:00 24:00

Times

Create New Time Schedule Period Module

3. Click **Times**. The occupation times (if any) for the exception are displayed:

Open Day Event: Times

Tech Pubs IQ422
Address 16 on Lan 18
Current User: Admin

Logout Alarms Time Schedules Modules GraphIQs

Address Page
Alarm Destinations
Alarm Groups
Alarm Routes
Analogue Nodes
Digital Bytes
Digital Inputs
Directories
Displays
Drivers
Functions
IC Comms
Interface Modules

Open Day Event

Time Value Delete

Create New Time Schedule Period Module

4. If there are no time periods set up, or to add another, click **Create New Time Schedule Period Module**.

Note: New periods should be always added in pairs, one to switch occupation on, and one to switch it off.

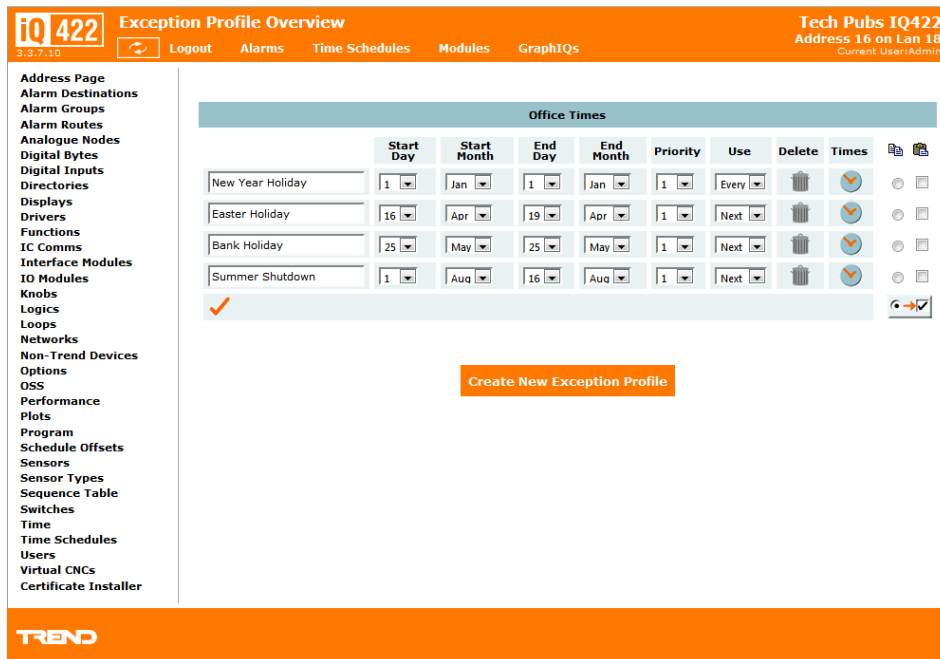
5. In the box in the **Time** column enter the required start (on) or stop (off) time, in the 24-hour format hours:minutes (e.g. 17:30 for 5.30pm). To change a time from a start time to a stop time or from stop time to a start time, select **On** or **Off** in the **Value** column.
6. To delete a time period, click . The controller will prompt for confirmation, click **OK**.
7. Click to save the new occupation times.

4.3.4.4 Copy Settings from an Exception

All of the parameters and occupation times for an exception can be copied to another. Any existing settings in the destination exception will be overwritten.

To copy exception settings:

1. View the exceptions (see [“View Exceptions” on page 19](#)).



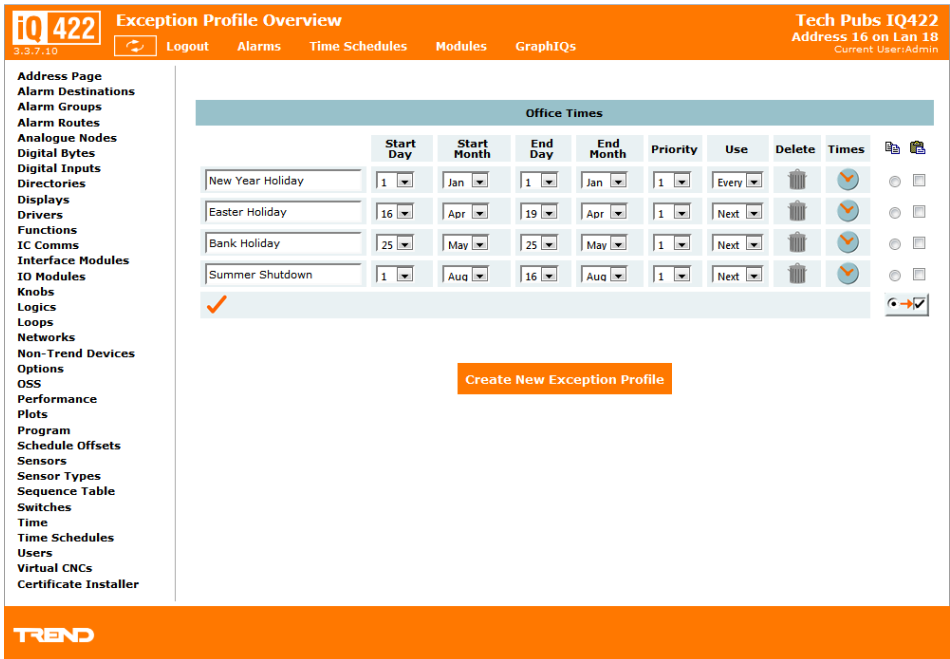
2. Select the radio button in the column next to the exception from which the settings are to be copied.
3. Select the check box(es) in the column next to the exception(s) to which the information is to be pasted.
4. Click to copy the settings to the selected exception(s).

4.3.4.5 Delete an Exception

Any exceptions that are no longer required can, and should, be deleted to free up memory available in the controller.

To delete an exception:

1. View the exceptions (see [“View Exceptions” on page 19](#)).



2. Click next to the exception to be deleted.
3. The controller will prompt for confirmation, click **OK**.

4.4 View and Adjust Module Parameters

Many module parameters can be viewed and adjusted, subject to your access rights. Typical examples are adjusting the value of a knob (e.g. to change a temperature setpoint) or a change the state of a switch (e.g. occupation override), both of which are described in the following pages.

For help with other module types, refer to [“Advanced Procedures” on page 31](#).

Note: Only the module's parameters can be changed; changes to the strategy structure (i.e. module interconnections, and creation/deletion of modules) are not allowed.

4.4.1 Adjust the Value of a Knob

The value of a knob can be adjusted providing you have the appropriate user level.

To adjust the value of a knob:

1. Login to the IQ4 Controller (see [“Login Procedure” on page 15](#)).
2. Click **Modules** on the menu bar.
3. Click **Knobs** in the navigation panel. A list of up to 25 knob modules will be displayed. If there are more than 25 modules, use the navigation controls to change the list view (see [page 33](#)).
4. Click the underlined module reference (e.g. K1) to display the module detail page:

Parameter	Value	Connection
Label	Preheat Set Point	
Units	DegC	
Value	12.00	
Destination	12.00	L10
User Level	99	
Top Range	20.00	
Bottom Range	10.00	
Clip To Limits	Off	
CDV Increment	0.25	
Sequence Table	0	
Page	2	
X Co-ordinate	57	
Y Co-ordinate	152	

5. In the **Value** box enter the new value for the knob.
6. Click  to send the changes to the controller.

4.4.2 Change the Status of a Switch

The status of a switch can be changed providing you have the appropriate user level.

To change the status of a switch:

1. Login to the IQ4 Controller (see [“Login Procedure” on page 15](#)).
2. Click **Modules** on the menu bar.
3. Click **Switches** in the navigation panel. A list of up to 25 switch modules will be displayed. If there are more than 25 modules, use the navigation controls to change the list view (see [page 33](#)).
4. Click the underlined module reference (e.g. W1) to display the module detail page.

Parameter	Value	Connection
Label	Occupation Override	
Status	On	
Destination	On	G1E
User Level	0	
Sequence Table	0	
Page	4	
X Co-ordinate	36	
Y Co-ordinate	150	

5. In the **Status** box select the required status (e.g. On or Off).
6. Click  to send the changes to the controller.

4.5 Display a Graph

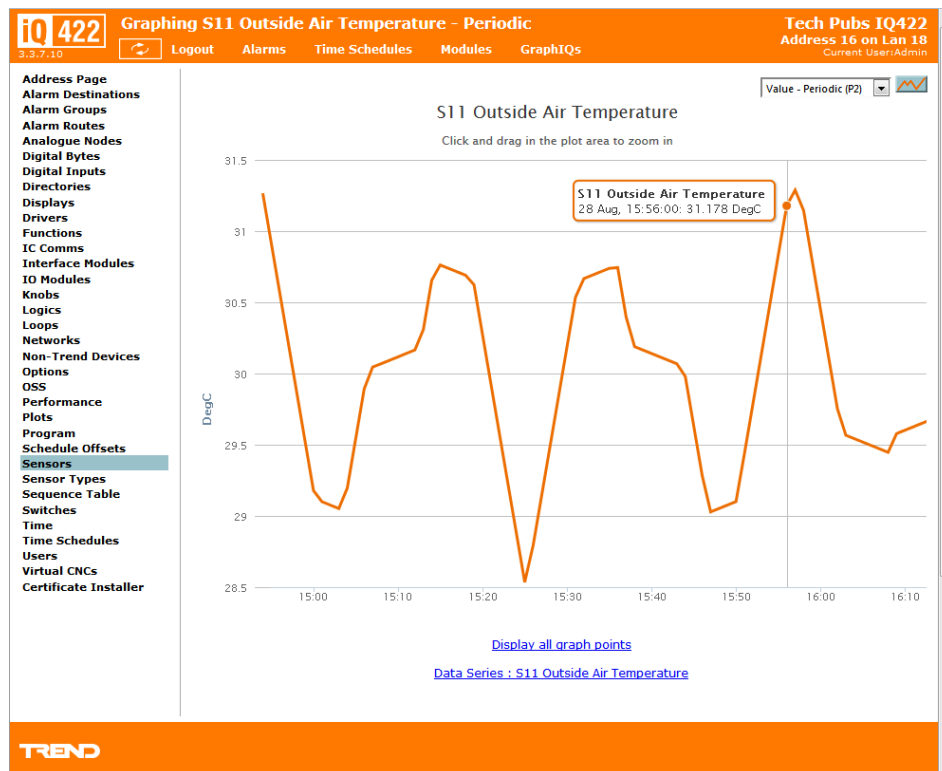
A graph can be displayed for any module value that is being logged in the controller (via a Plot module). Any module with a graph icon  next to it can display a graph.

To display a graph of data logged in the controller:

1. Login to the IQ4 Controller (see [“Login Procedure” on page 15](#)).
2. Click **Modules** on the menu bar.
3. Click the required module type in the navigation panel.

Note: View the Plots module list to see all modules that can be viewed as a graph.

4. Click  for the required module. A graph will be displayed generated from up to 100 of the most recently logged values (see [“Show More or Less Graph Points”](#) below):



5. To display the measured value for a specific point on the graph, move the mouse pointer over the plot trace until a vertical line appears. Where the line meets the trace, a box will appear showing the date, time and measured value of that point on the graph.
6. If there is more than one graph available for the selected module (e.g. different type or interval) select the required graph using the list box in the top right corner, then click .

4.5.1 Show More or Less Graph Points

A graph is initially displayed showing up to 100 of the most recently logged values. Up to 1000 values can be stored in the controller.

To re-display the graph with up to 1000 points:

1. Click **Display all graph points** below the graph.

To re-display the graph with up to 100 points:

1. Click  on the menu bar

4.5.2 Display the Graph Data

The source data used to create a graph can be displayed as a list of dates, times and measured values.

To display the graph data:

1. Ensure that the required graph is displayed (see [page 27](#)).
2. Click the **Data Series** link below the graph. A list of logged data is displayed:

Data Series : S11 Outside Air Temperature					
Tue,	28	Aug	2012	14:57:00,	30.222
Tue,	28	Aug	2012	14:58:00,	29.874001
Tue,	28	Aug	2012	14:59:00,	29.525999
Tue,	28	Aug	2012	15:00:00,	29.178001
Tue,	28	Aug	2012	15:01:00,	29.1
Tue,	28	Aug	2012	15:02:00,	29.076
Tue,	28	Aug	2012	15:03:00,	29.052002
Tue,	28	Aug	2012	15:04:00,	29.195402
Tue,	28	Aug	2012	15:05:00,	29.543402
Tue,	28	Aug	2012	15:06:00,	29.891399
Tue,	28	Aug	2012	15:07:00,	30.045
Tue,	28	Aug	2012	15:08:00,	30.069
Tue,	28	Aug	2012	15:09:00,	30.093
Tue,	28	Aug	2012	15:10:00,	30.116999
Tue,	28	Aug	2012	15:11:00,	30.141001
Tue,	28	Aug	2012	15:12:00,	30.165001
Tue,	28	Aug	2012	15:13:00,	30.307798
Tue,	28	Aug	2012	15:14:00,	30.655796
Tue,	28	Aug	2012	15:15:00,	30.761997
Tue,	28	Aug	2012	15:16:00,	30.737999
Tue,	28	Aug	2012	15:17:00,	30.714001
Tue,	28	Aug	2012	15:18:00,	30.690001
Tue,	28	Aug	2012	15:19:00,	30.622799
Tue,	28	Aug	2012	15:20:00,	30.274801
Tue,	28	Aug	2012	15:21:00,	29.9268
Tue,	28	Aug	2012	15:22:00,	29.578798
Tue,	28	Aug	2012	15:23:00,	29.230801
Tue,	28	Aug	2012	15:24:00,	28.882801
Tue,	28	Aug	2012	15:25:00,	28.534801
Tue,	28	Aug	2012	15:26:00,	28.794601
Tue,	28	Aug	2012	15:27:00,	29.142601
Tue,	28	Aug	2012	15:28:00,	29.490599
Tue,	28	Aug	2012	15:29:00,	29.8386

*Note: The list will contain up to 100 of the most recent values used to generate the graph (or up to 1000 values if **Display all graph points** was selected).*

3. To copy some or all of the listed values to use in another software application, drag the mouse cursor to highlight the range of values required, then copy to the clipboard (e.g. by pressing CTRL+C).
4. Click  to close the list.

4.5.3 Zoom in and Out of a Graph

It is possible to zoom in and out on any graph so that the detail can be more easily seen.

To zoom in on the graph:

1. Display the required graph (see [page 27](#)).
2. Click the mouse button and drag over the required area of the graph.
3. Release the mouse button. The graph will zoom to the selected area.

To return to the whole graph:

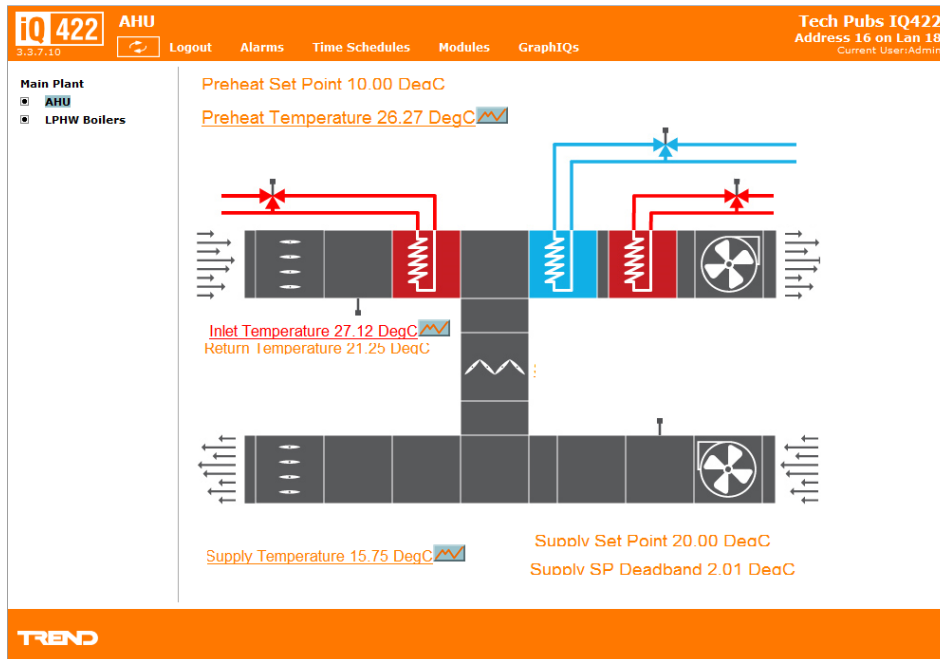
1. Click the **Reset Zoom** button.

4.6 View Graphical Display Pages (GraphIQs)

If they have been configured on the controller GraphIQs provide a graphical way of viewing information from the IQ4 controller, and making changes. If your user home page has been set up to do so, a GraphIQ may be displayed when you login.

To view graphical display pages:

1. Login to the IQ4 Controller (see [“Login Procedure” on page 15](#)).
2. Click **GraphIQs** on the menu bar. The top-level GraphIQ page is displayed.



The navigator panel on the left shows all of the directory modules in the controller in a hierarchical/tree structure. Selecting a directory module causes the associated display modules to be displayed in the information panel.

Note: Depending on your user profile, the navigator panel may not be displayed.

3. If the navigator panel is displayed, choose the required page by clicking the directory tree and then clicking the page name.

If the navigator is not displayed, use the links provided in the displayed page to navigate to other GraphIQs pages.

4. The information displayed will depend on how the GraphIQ page has been engineered. In general any underlined text will either open a Module Detail Page for the displayed parameter, or navigate to a different GraphIQ page.

5 ADVANCED PROCEDURES

This section covers less commonly used procedures, typically required by system administrators.

5.1 View and Adjust Module Parameters (General Guidance)

Many module parameters can be adjusted, subject to your access rights. Changes to the strategy structure (i.e. module interconnections, and creation/deletion of modules) are not possible using the web interface. However, it is possible to see and follow the interconnections between modules.

To adjust a module's configuration parameters:

1. Login to the IQ4 Controller (see [“Login Procedure” on page 15](#)).
2. Click **Modules** on the menu bar.
3. Click the required module type in the navigation panel.

Note: The module types listed are dependent on your user level and profile.

The display will change to display a Module Overview which will list basic information for up to 25 modules. If there are more than 25 modules, use the navigation controls to change the list view (see [page 33](#)).

The information displayed will vary depending on module type but will typically include Item (module number), Label (descriptive name), and Value. The following shows an example list of Sensor modules:

Item	Label	Value	Units	Graph	Alarm
S1	Room Temp	21.45	DegC	Value - 1 Minute (P4) Value - Periodic (P5)	No Alarm
S2	External Digital 2	3.87	DegC	Value - 1 Second (P1)	No Alarm
S3	Inlet Temperature	30.26	DegC	Value - 1 Second (P6) Value - Change of Value (P8)	No Alarm
S4	Sensor 1	17.05	DegC		No Alarm
S5	Internal Digital	1.49			

If a module has a button next to it, a graph of recent values can be viewed (see [“Display a Graph” on page 27](#)).

Any modules that have a current alarm condition associated with them are displayed in red.

4. Click the underlined module number to display the Module Details Page, e.g. S3 to show Sensor 3:

IQ 422

3.3.7.10

Sensor 3 - Inlet Temperature

Login Alarms Time Zones Modules GraphIQs

Tech Pubs IQ422

Address 16 on Lan 18

Current User:Admin

Address Page

Alarm Destinations

Alarm Groups

Alarm Routes

Analogue Nodes

Digital Bytes

Digital Inputs

Directories

Displays

Drivers

Functions

IC Comms

Interface Modules

IO Modules

Knobs

Loops

Networks

Non-Trend Devices

Options

OSS

Performance

Plots

Program

Schedule Offsets

Sensors

Sensor Types

Sequence Table

Switches

Time

Time Schedules

Users

Virtual CNCs

Certificate Installer

Parameter Value Connection

Label Inlet Temperature

Disable Module No

Input Type 1: Internal Analog

Override Output No

Override Value 20.00

Units DegC

Source(mV) 30.26 F29D

Value 30.26 F5E

Type 0

Offset 24.00

High Alarm Level 12.00

Low Alarm Level 0.00

Read 0

Out of Limits 0

Low 0

High I

In Alarm High

Alarm State 0001

For most types of module the information displayed will include:

- a list of parameters,
- the current value for each parameter, and
- references to any other connected module (see [“View a Connected Module” on page 33](#)).

The parameters displayed will vary depending on the type of module and your user level. If your user level is high enough, you can make changes to certain module parameters.

5. Clicking  at the top of the page will display the parameters for next module, and clicking  will display the parameters for the previous module.
6. If required, enter a new value in the **Value** column for the parameter(s) to be adjusted.
7. Once any required changes have been made click  to send an update to the controller.

5.1.1 Navigation Controls (Module List)

The Module List Page will show a maximum of 25 modules at one time. Where there are more than 25 modules of the same type you can scroll up and down using the navigation controls at bottom of the list. If you know the number of a module, you can also jump directly to that position in the Module Page listing.

To display other modules:

1. Click the appropriate navigation button:

<i>Button</i>	<i>Function</i>
	First 25 modules
	Next 25 modules
	Previous 25 modules
	Last 25 modules

To display the next 25 modules from a specified module:

1. In the box next to **Get From** enter the required module (e.g. for Knob K23, type in **23**).
2. Click **Get From**.

5.1.2 View a Connected Module

Where the module is connected to another module (in the controller strategy), this will be indicated in the **Connection** column. You can go directly to a module that is a source to, or destination of the current module. This enables you to move easily through the strategy.

Note: No structural changes can be made to the strategy using the web interface.

To go to the connected module:

1. Where more than one module is connected, you can choose the required module using the pick list in the **Connection** column.
2. Click  to go to the destination module (i.e. this parameter value is being used by the connected module), or click  to go to the source module (i.e. this parameter value is being set by the connected module).

5.2 Add a User

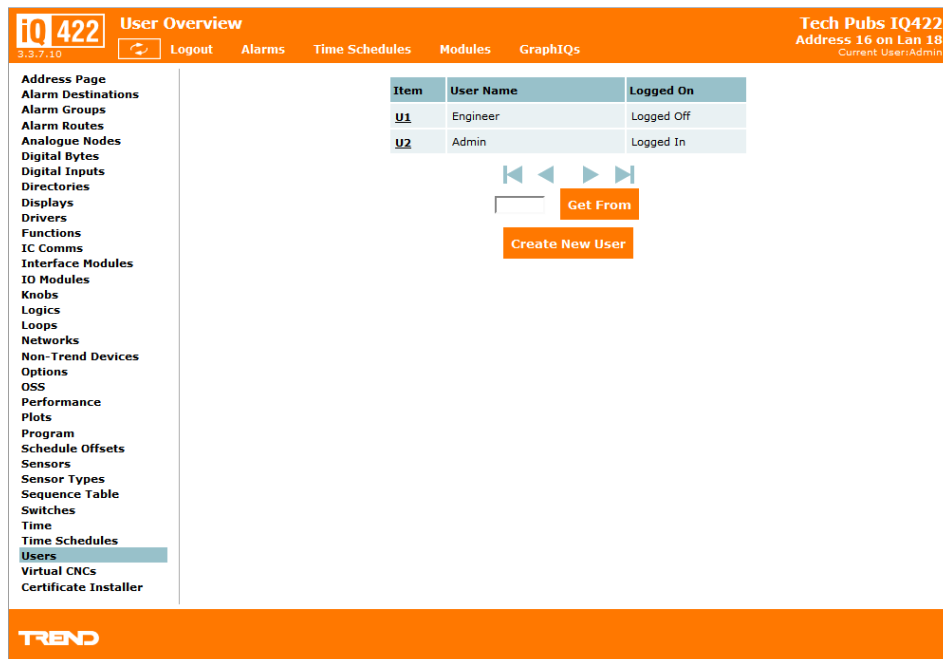
If required additional users can be set up. Any user added via the web interface will be given ‘web only’ access to the controller, i.e. they will not be able to make changes using text communications (text comms). If the ability to make changes using text comms is required, then the user must be created using SET.

To add a user:

1. Login to the IQ4 Controller (see [“Login Procedure” on page 15](#)).

Note: You must have a user level of at least 99.

2. Click **Modules** on the menu bar.
3. Click the **Users** module type on the left of the screen. The **User Overview Page** will appear.



4. Click **Create New User**. The following box is displayed.

The 'Create New : User' dialog box is shown. It has three input fields: 'User Name', 'Password', and 'Re-type Password'. At the bottom are 'OK' and 'Cancel' buttons.

5. In the **User Name** box enter the required user name. This is a name (of up to 30 characters) used when logging in from a web browser and can comprise any ASCII characters except \ / ({ ; , :
6. In the **Password** box enter the password associated with the user name. This must be entered by the user when logging in from a web browser and can comprise any ASCII characters.
7. In the **Re-type Password** box re-enter the password typed in step 7.
8. Click **OK**. The new user will be added with the following default settings:

Parameter	Value
User Level	100
Home Page	(not specified)
Timeout (Minutes)	15
View Alarms, Time Schedules, Modules, GraphIQs, Navigator	All On

9. Edit the new user module as required see [“Change User Settings” on page 35](#).

5.3 Change User Settings

Certain user parameters can be changed using the web interface, including user level, timeout, options displayed on the menu bar and whether or not the GraphIQs navigator is displayed.

Note: Users cannot be deleted using the web interface; they can only be deleted using SET.

To edit a user:

1. Login to the IQ4 Controller (see [“Login Procedure” on page 15](#)).

Note: You must have a user level of at least 99.

2. Click **Modules** on the menu bar.
3. Click the **Users** module type on the left of the screen. The **User Overview Page** will appear.
4. Click the user module reference number in the **Item** column. A detail page for the user will appear:

Parameter	Value	Connection
User Name	Engineer	
Password	*****	
PIN	*****	
Level	100	
Home Page		
Timeout (Minutes)	15	
Logged On	Logged Off	
View Alarms	On	
View Time Schedules	On	
View Modules	On	
View GraphIQs	On	
View Navigator	On	
Page	0	
X Co-ordinate	0	
Y Co-ordinate	0	

5. Click on **User Name** or **Password** to open a separate dialogue box to change these settings:

Adjust : User

User Name:

Password:

Re-type Password:

OK Cancel

6. Type the new user name or password as required and click **OK**.
7. Change the values of any other parameters as required.
8. Click to send the changes to the controller.

INDEX

A

About

IQ4 controller range	7
this manual	5

Access Rights	12
---------------------	--------------------

Alarm Log	30
-----------------	--------------------

Alarms Page	30
-------------------	--------------------

C

Connecting to an IQ4 Controller	9
---------------------------------------	-------------------

Contacting Trend.....	6
-----------------------	-------------------

Controller

address	11
firmware version	10
name (identifier)	11
type	10

Conventions Used in this Manual	5
---------------------------------------	-------------------

D

Destination Module	33
--------------------------	--------------------

Display Refresh	12
-----------------------	--------------------

E

Exceptions

adding	20
change parameters	21
change times	22
copy settings from	23
deleting	24
viewing	19

G

Goto Connected Module	33
-----------------------------	--------------------

Graph Data	28
------------------	--------------------

Graphical Display Pages

viewing	29
---------------	--------------------

GraphIQs. See Graphical Display Pages

Graph Pages

displaying	27
show more/less points	27
view source data	28
zooming in/out	28

Guest Users	12
-------------------	--------------------

I

IQ4 Controller

connecting to	9
overview	7

K

Knob

adjusting value	25
-----------------------	--------------------

L

Login Procedure	15
-----------------------	--------------------

Logout Procedure	15
------------------------	--------------------

M

Menu Bar

location of	10
overview	11
user settings	35

Module List Pages

viewing	25
---------------	--------------------

Module Parameters

changing	31
knob value	25
switch status	26
user settings	35
viewing connected modules	33

N

Navigation Controls

connected modules	33
menu bar	11
module lists	33

Navigator Panel

hiding from users	35
overview	10, 11

O

Occupation Times. See also Exceptions

changing	18
copying	18
viewing	17

P

Pages. See Web Pages

Port Number	9
-------------------	-------------------

R

Refresh the Display	12
---------------------------	--------------------

S

Saving Changes	12
----------------------	--------------------

Security	12
----------------	--------------------

Source Module	33
---------------------	--------------------

Support Contact Details	6
-------------------------------	-------------------

Switch

changing status	26
-----------------------	--------------------

System Guest User	12
-------------------------	--------------------

T

Technical Support	6
-------------------------	-------------------

Time Schedule Overview Page

changing normal occupation times	18
creating or changing exceptions	20
viewing exceptions	19
viewing occupation times	16

Title Bar	10
-----------------	--------------------

U

Users

access rights	12
adding	34
change settings	35
current	11
guest	12
logging in	15
logging out	15
system guest	12

V

Value Adjustment	31
------------------------	--------------------

W

Web Browser

supported types	9
-----------------------	-------------------

Web Pages

Alarms Page	30
GraphIQ Page	29
layouts on different devices	13
Welcome Page	10

Welcome Page	10
--------------------	--------------------

Z

Zooming In or Out of Graphs	28
-----------------------------------	--------------------

Trend Control Systems Limited

St. Mark's Court, North Street, Horsham, West Sussex, RH12 1BW, UK. Tel: +44 (0)1403 211888, www.trendcontrols.com
